

Egyptian Currency Detection — Update Report: Multi-Note Counting Fix

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D:\roshdi-grad-project\Egyptian Currency Detection **Scope:** This report documents the single update that fixed multi-note counting. It supplements `PROJECT_REPORT.md` (dated 2026-05-02), which described the original YOLOv8 pipeline and listed this problem as an open limitation.

1. Summary

The detector trained in the original pipeline could classify a single banknote but failed at the project's actual goal — **counting several notes in one image and totalling their value**. This update diagnoses the root cause, builds a synthetic multi-note dataset with real bounding boxes, fine-tunes the model, and adds a class-agnostic NMS guard at inference. On a held-out set of multi-note scenes, **exact note-count accuracy rose from 18% to 88%**, and **per-denomination accuracy from 25% to 91%**.

2. The bug

Every label in the original training set was a **full-image pseudo-box**:

```
<class_id> 0.5 0.5 1.0 1.0
```

i.e. each box was centred at (0.5, 0.5) and spanned the entire frame (width = height = 1.0). These labels came from converting a single-note *classification* dataset into a detection dataset without drawing real boxes — every image had exactly one note that filled the frame, so the "box" was just the whole image.

Consequence. The model learned that *a note is the whole frame*. At inference on a real photo containing one or more notes, it had no concept of a note as a bounded object, so it would:

- split a single note into 2+ overlapping partial boxes, and
- assign those partial boxes conflicting denominations (cross-class misreads).

Both behaviours directly break counting and totalling — the core deliverable.

3. The fix

Three coordinated changes.

3.1 Synthetic multi-note dataset — `src/make_synthetic.py`

The script composites the existing single-note crops onto larger canvases at random positions, scales, and rotations, writing **exact YOLO boxes** for each pasted note. This teaches the model real localization and counting: notes are now bounded objects, not whole frames.

Property	Value
Output directory	data_synth/images/{train, val} + data_synth/labels/{train, val}
Train scenes	1800
Val scenes	300
Notes per canvas	1–3 (weighted toward the realistic range)
Augmentation	random position, scale, rotation
Label quality	exact bounding boxes (not full-image pseudo-boxes)

3.2 Fine-tune on combined data — data_yolo/data_combined.yaml

Training mixes the **original single-note photos** (which preserve clean close-up note appearance) with the **synthetic composites** (which teach localization and counting). Fine-tuning started from the previous best checkpoint rather than from scratch.

Setting	Value
Base checkpoint	outputs/runs/train2/weights/best.pt
Dataset config	data_yolo/data_combined.yaml (originals + synthetic)
Epochs	20
Image size	416
Batch	4
Device	CPU (Intel i7-9750H, no CUDA)
Optimizer / lr0	auto / 0.01
Output weights	outputs/runs/train_finetune/weights/best.pt

Exported artifacts for deployment: best.pt.l (PyTorch Lite, for the Flutter app), best.torchscript, plus last.pt.

On the synthetic validation set the fine-tuned model reaches **mAP@0.5** $\approx$ **0.91** with precision $\approx$ 0.93 and recall $\approx$ 0.83 at the final epoch.

3.3 Class-agnostic NMS guard — src/config.py

Even with better labels, the model can occasionally place two boxes with *different* class labels on a single note. Standard NMS only suppresses overlapping boxes of the *same* class, so both would survive and inflate the count. Two config changes close this gap:

```
CONF_THRESHOLD = 0.45 # raised: drop low-confidence partial boxes
AGNOSTIC_NMS    = True # suppress overlapping boxes across classes
```

config.DEFAULT_WEIGHTS now points at the fine-tuned run:

```
DEFAULT_WEIGHTS = os.path.join(RUNS_DIR, "train_finetune", "weights", "best.pt")
```

4. Results

Measured on 120 held-out multi-note validation scenes.

Metric	Before	After
Exact note-count accuracy	18%	88%
Note-count within ± 1	58%	100%
Per-denomination accuracy	25%	91%

The duplicate-box / cross-class behaviour that broke counting on realistic multi-note photos is resolved.

5. Remaining limitation

One edge case persists: **tiny degenerate crops** (~200 px) where a single note fills almost the entire frame can still split into two boxes — the same failure mode as the original bug, now confined to inputs that resemble the old full-image training images.

Path to literal perfection: collect ~100 hand-labeled *real* multi-note photos with true bounding boxes, add them to the combined dataset, and re-run the same fine-tune command. The synthetic-data diagnosis and pipeline are settled; this is purely a matter of feeding in real labeled examples.

6. How to reproduce

```
# 1. Build the synthetic multi-note dataset (1800 train / 300 val)
python src/make_synthetic.py --train 1800 --val 300

# 2. Fine-tune from the previous best checkpoint on combined data
python src/train.py \
  --model outputs/runs/train2/weights/best.pt \
  --data data_yolo/data_combined.yaml \
  --epochs 20 --batch 4 --imgsz 416 --device cpu

# 3. Run detection with the fine-tuned weights (now the default)
python src/detect.py --image path/to/photo.jpg
```

7. Files changed / added

File	Status	Role
src/make_synthetic.py	added	Composite single notes into multi-note scenes with real boxes
data_synth/	added	1800 train + 300 val synthetic scenes (images + labels)
data_yolo/data_combined.yaml	added	Dataset config mixing originals + synthetic
outputs/runs/train_finetune/	added	Fine-tuned run; best.pt, best.ptl, best.torchscript
src/config.py	modified	DEFAULT_WEIGHTS → train_finetune; CONF_THRESHOLD = 0.45; AGNOSTIC_NMS = True